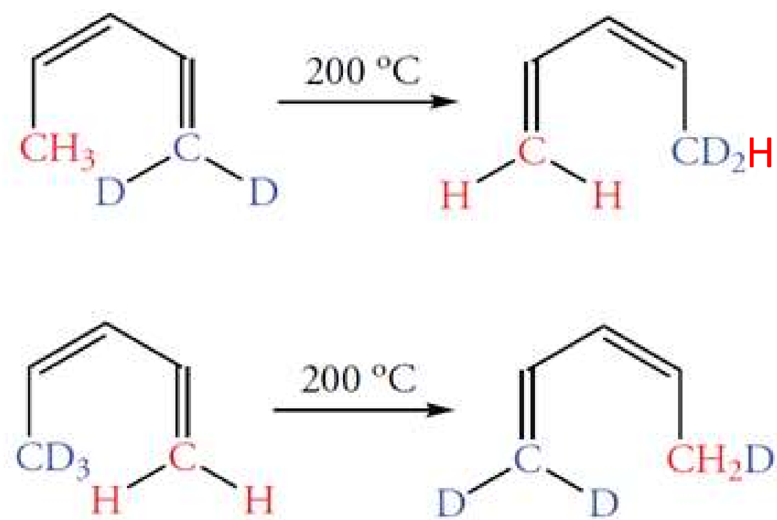
# Sigmatropic Rearrangements

- **Sigmatropic rearrangements** are intramolecular reactions in which one atom or group of atoms linked by a $\sigma$ bond migrates from one end of a $\pi$ system to the other. In the process, the positions of single and double bonds simultaneously shift.

- Examples of one type of sigmatropic rearrangement are the isomerizations of the following deuterated 1,3-pentadienes.

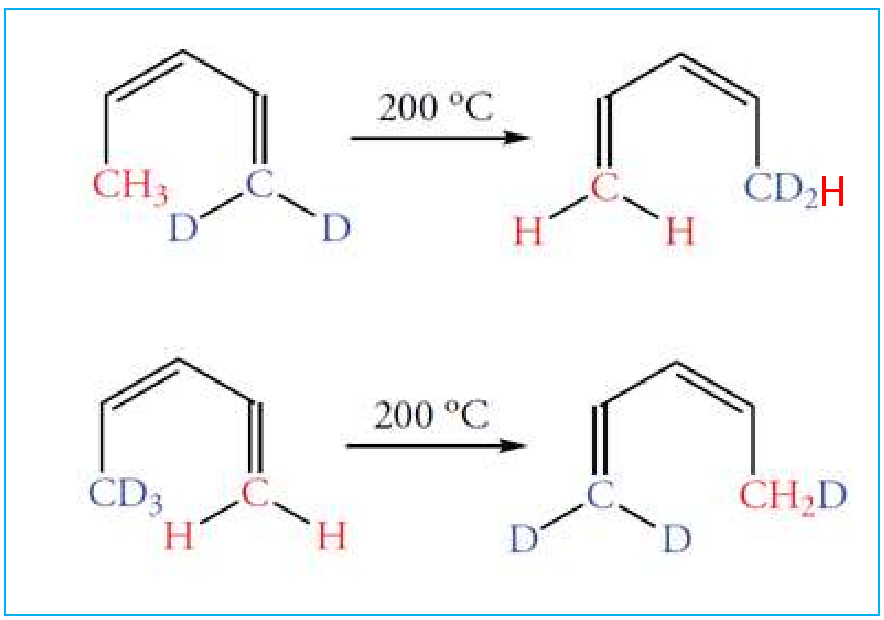
sigma designation from single carbon–carbon/other atom bonds and the Greek word *tropos*, meaning turn.

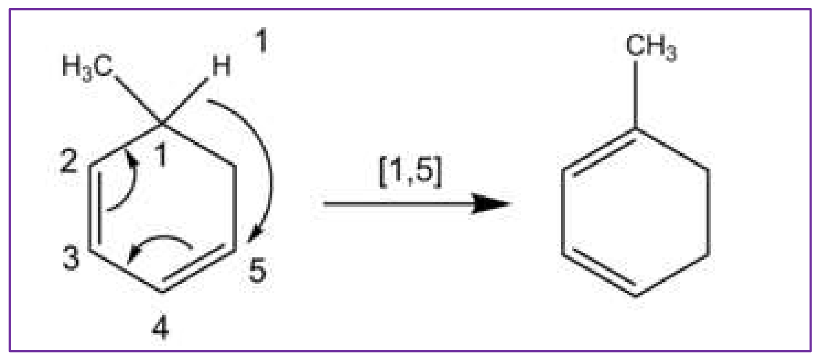



In these two rearrangements, a hydrogen (or deuterium) atom that is a bonded at an $sp^3$-hybridized carbon atom moves from one end of a five-carbon-atom system to the other end. In the process, the locations of the alternating single and double bonds and the hybridization of the terminal atoms change.

# Sigmatropic Rearrangements

- Sigmatropic rearrangements are identified by two numbers separated by a comma and enclosed within brackets.
- The two numbers refer to the number of atoms in the two groups connected by a σ bond. The numbers are also related to the orbitals in each piece of the original molecule that are involved in the rearrangement.
- The rearrangement of the 1,3-pentadiene occurs by a shift of a hydrogen atom across a five-carbon-atom system and is therefore a [1,5] sigmatropic rearrangement.
- The number "1" refers to the orbital of the hydrogen atom, which is bonded to one end of the system that rearranges. The number "5" refers to the five carbon 2p orbitals of the reacting system.



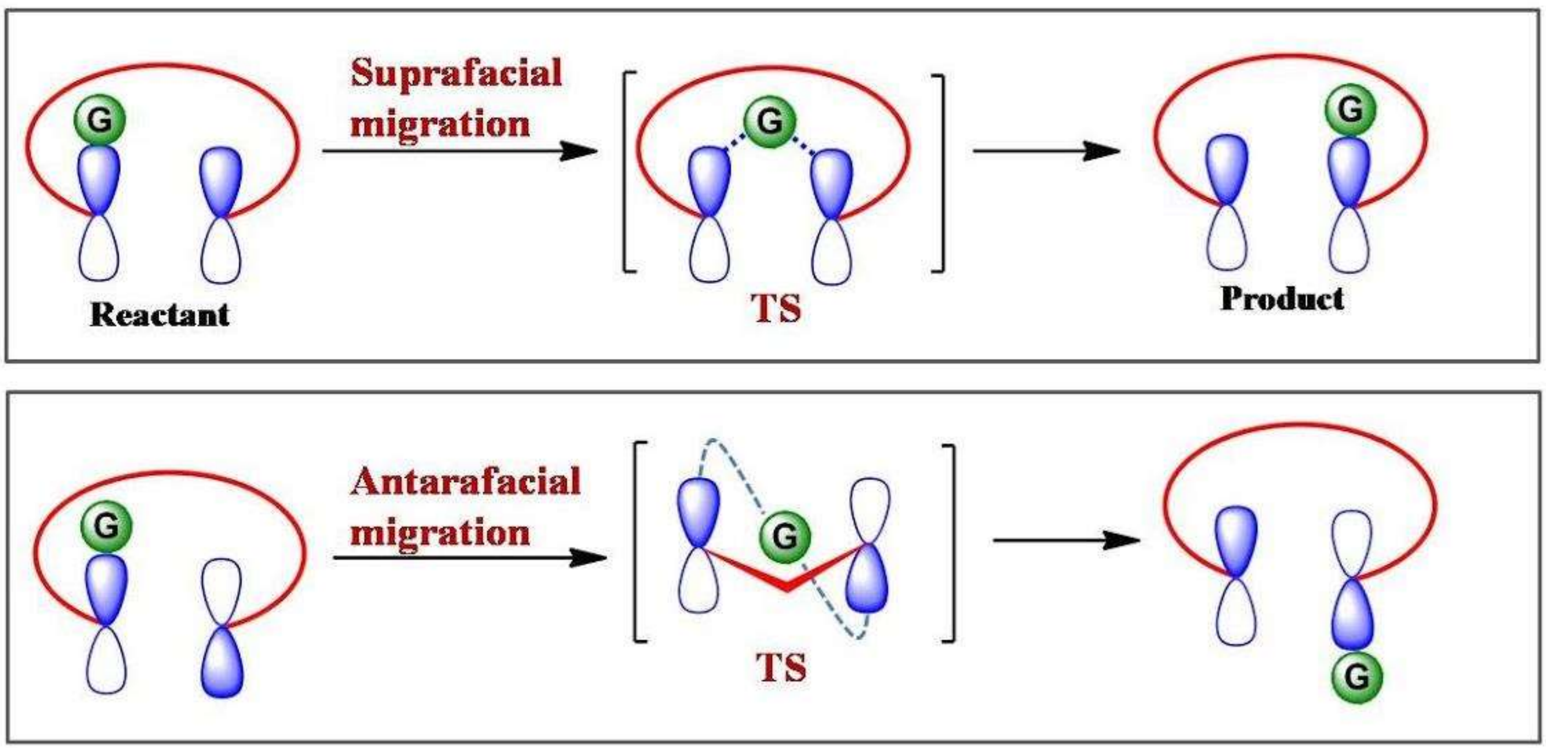
# Sigmatropic Rearrangements

The migration of a group from one site of a π system is called a sigmatropic rearrangement. This migration can occur in two ways.

1. Migration across the same face is a **suprafacial rearrangement**.

2. Migration from one face to the other face is an **antarafacial rearrangement**.

Therefore, sigmatropic rearrangements are controlled by the **same orbital symmetry considerations as cycloaddition reactions**.



# Molecular Orbitals in Sigmatropic Rearrangements

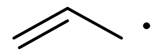
- To analyze sigmatropic rearrangements, it is convenient to consider a homolytic cleavage of the $\sigma$ **bond of the migrating groups**.

- Two radical fragments result from homolytic cleavage, and we can then examine the symmetry of their individual molecular orbitals. The reaction does not occur by this mechanism, but the stereochemical consequences are the same if the "radicals" are not allowed to move out of contact with each other.

- We will consider the **allyl system** to explain **[3,3] sigmatropic** rearrangements and the **pentadienyl system** to explain **[1,5] sigmatropic** rearrangements, we will first examine the molecular orbitals for these two systems.

# Molecular Orbitals in Sigmatropic Rearrangements

## Molecular Orbitals of an Allyl Radical



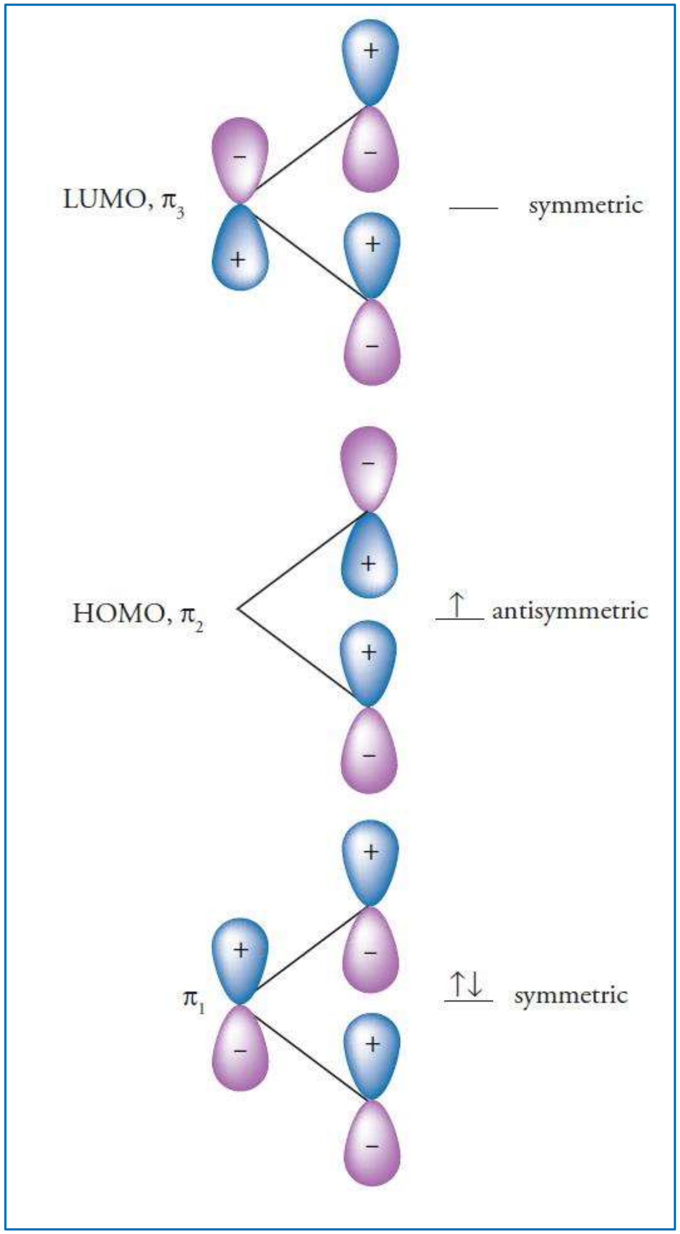
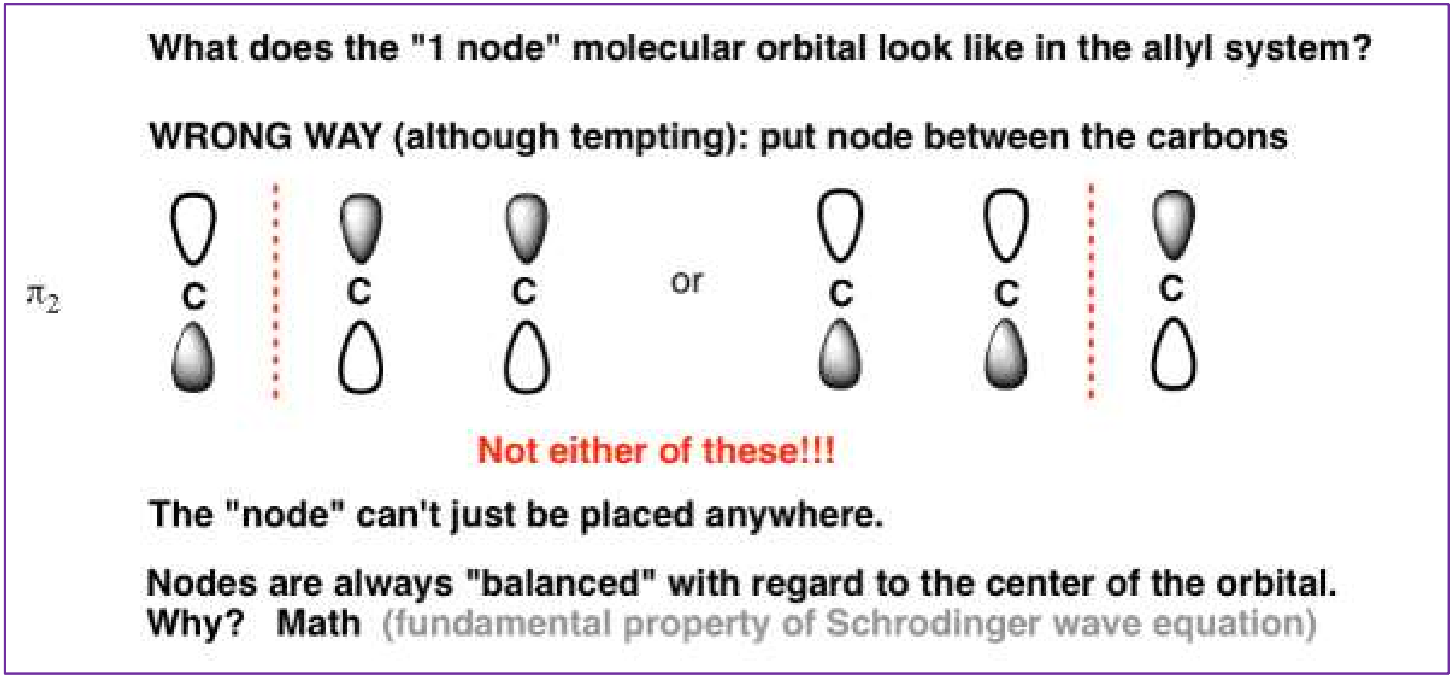



Figure shows the three **π orbitals** of the **allyl system**. The lowest energy MO is bonding over all three atoms and is symmetric.

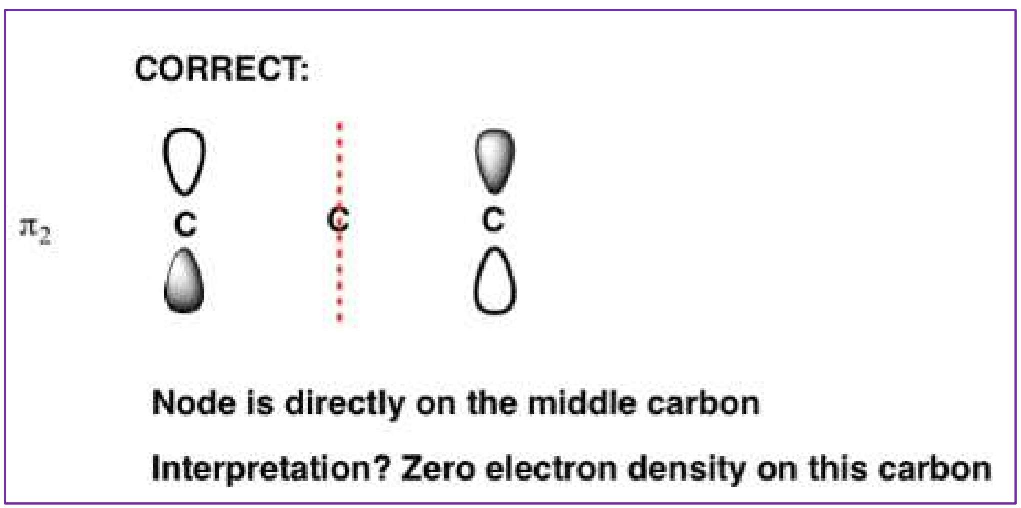
The **MO-labeled** $\pi_3$ is antibonding, has two nodes, and is symmetric.

The **MO-labeled** $\pi_2$ is nonbonding. Because there are three atoms, the single nodal plane of this MO must contain the center carbon atom. This MO, which is antisymmetric, is the one that participates in [3,3] sigmatropic rearrangements.

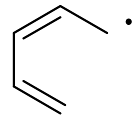
# The Tricky Middle Orbital Of The Allyl System




It turns out that because of the way the math works in the Schrodinger equation, the nodes are always “balanced” with respect to the centre of the orbital. The node can’t just be placed anywhere; when there is one node, it **must** be in the centre. [By the way, this is a general feature of MO’s where the number of contributing orbitals is odd, e.g. N=3, N=5, N=7… the first node will be on the central carbon.



## Molecular Orbitals of a Pentadienyl Radical



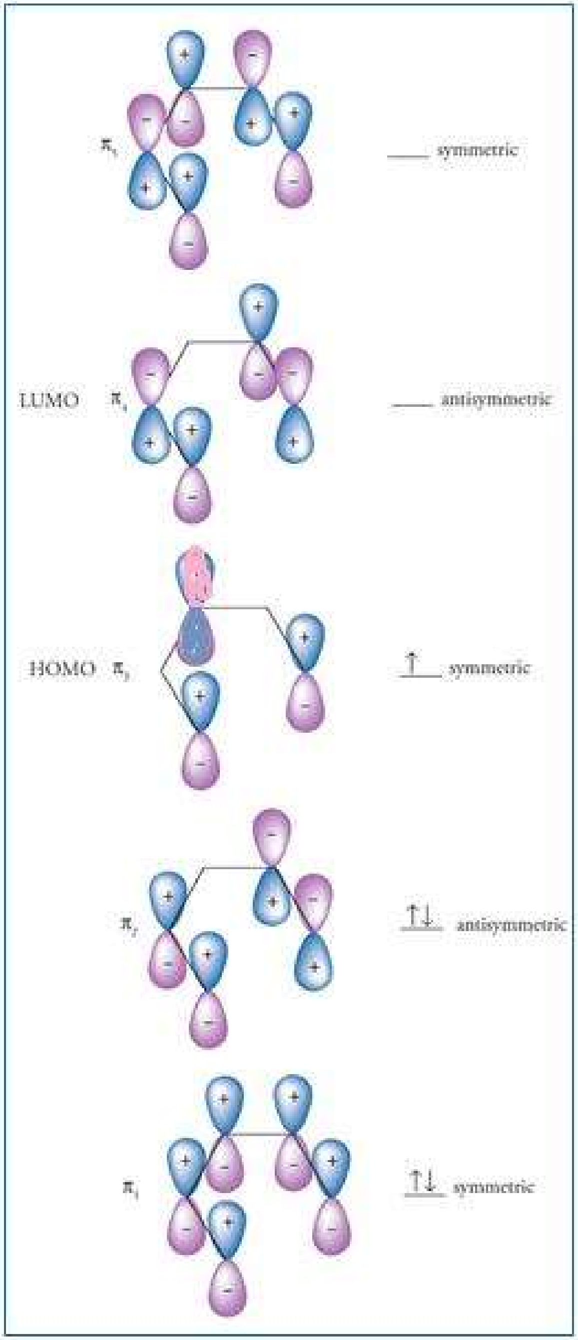



Figure shows the five π orbitals of the **pentadienyl system.** The lowest energy MO is bonding over all five atoms and is symmetric. The symmetry of each succeeding MO of higher energy alternates.

The MO required to explain [1,5] sigmatropic rearrangements is $\pi_3$.. It is symmetric.

# [1,5] Sigmatropic Rearrangements

The migration of a hydrogen atom over a pentadienyl system is the most common example of a [1,5] sigmatropic rearrangement.

$$\underset{1}{\overset{H \atop |}{C}H_2}-\underset{2}{CH}=\underset{3}{CH}-\underset{4}{CH}=\underset{5}{CD_2} \longrightarrow \underset{1}{CH_2}=\underset{2}{CH}-\underset{3}{CH}=\underset{4}{CH}-\underset{5}{\overset{H \atop |}{C}D_2}$$

If we *imagine* a homolytic cleavage of a C—H bond, the result is a hydrogen atom and a pentadienyl radical.

$$\underset{1}{\overset{H \atop \bullet}{\dot{C}}H_2}-\underset{2}{CH}=\underset{3}{CH}-\underset{4}{CH}=\underset{5}{CD_2}$$

The pentadienyl radical has five $\pi$ electrons that must be accommodated in the lowest energy molecular orbitals. Two electrons are located in the $\pi_1$ and $\pi_2$ molecular orbitals. The fifth electron is located in $\pi_3$, which is thus the HOMO. This molecular orbital is involved in the transfer of the migrating hydrogen atom. Because $\pi_3$ is symmetric, the terminal 2p orbitals have the same signs of the wave function on one side of the plane. As a result, the hydrogen atom moves across the surface of this orbital between C-1 and C-5 in a suprafacial process

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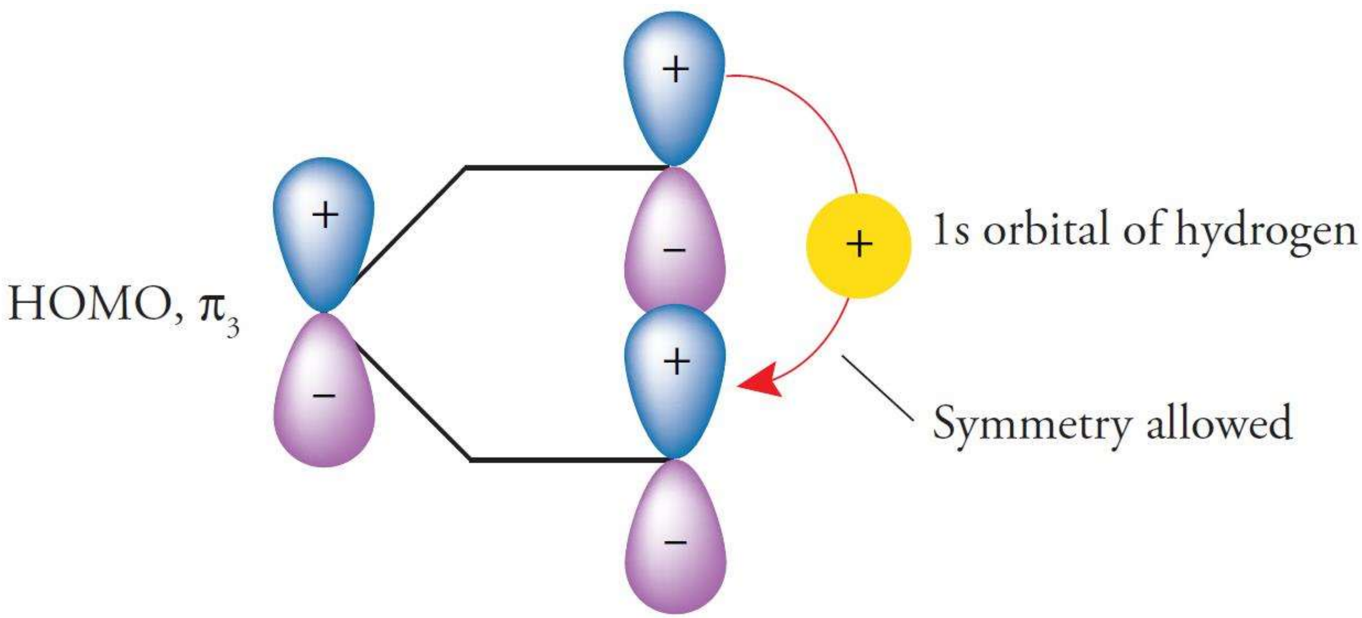
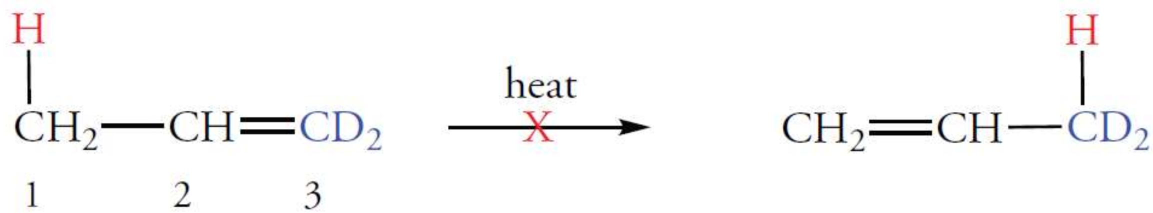
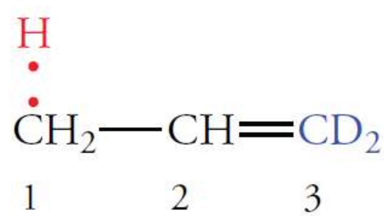


# [1,3] Sigmatropic Rearrangement

Sigmatropic rearrangements of a [1,3] type do not occur thermally. When we examine the molecular orbitals that would have to be involved in such a process, we can see why the model predicts that the reaction should not occur.

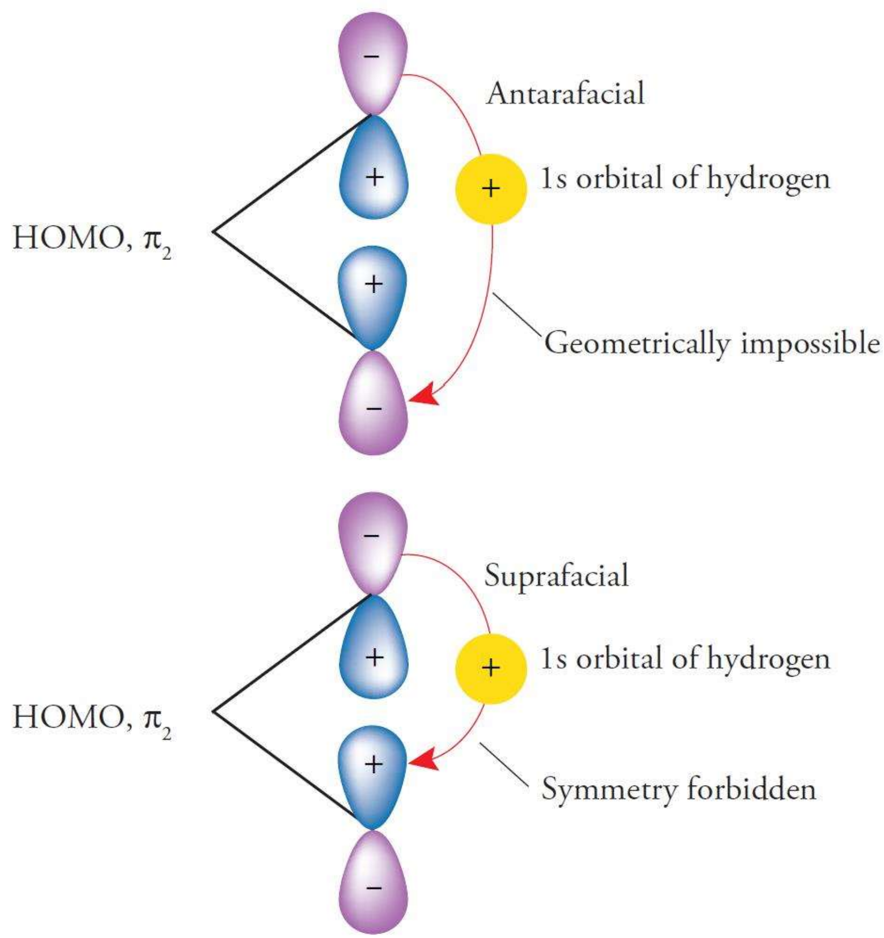



To visualize the reaction, we consider the homolytic cleavage of the C—H bond of the allyl system. Although the reaction does not occur this way, the orbitals involved in the concerted reactions have the same symmetry as those in the separate radical fragments. Thus, we consider the allyl radical and a hydrogen atom.

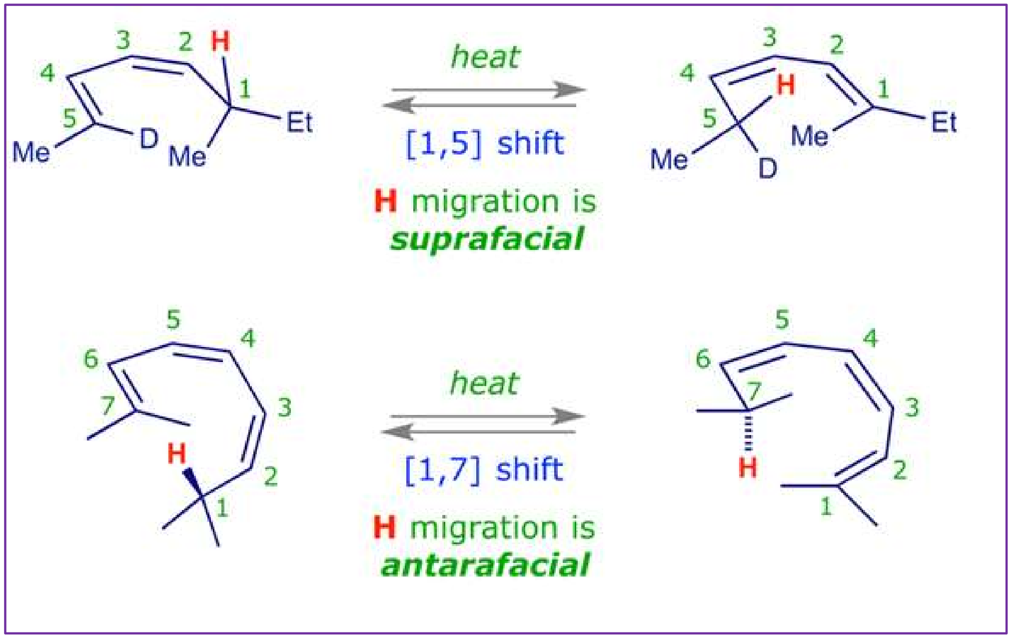


# [1,3] Sigmatropic Rearrangement

Two $\pi$ electrons occupy $\pi_1$ and one occupies the $\pi_2$ molecular orbital. Thus, the HOMO is $\pi_2$, which is antisymmetric. Migration of a hydrogen atom suprafacially is geometrically possible, but it is *not* allowed by orbital symmetry. The hydrogen atom cannot leave C-1 and bond to C-3 because the symmetries of the orbitals are different



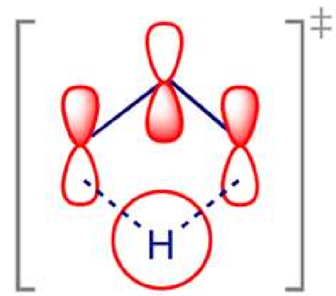
# Thermal [1,5] and [1,7] sigmatropic hydrogen shifts



# Photochemical [1,3] sigmatropic hydrogen shifts

This reaction is observed, and again we can analyse the FMOs to see how it works.


We view the reaction as the *cycloaddition of a proton to an allyl anion*, choosing the LUMO of $H^+$ (little choice here) and the HOMO of allyl anion ($\Psi_3$ in the photochemically excited state)



Consider the anion:

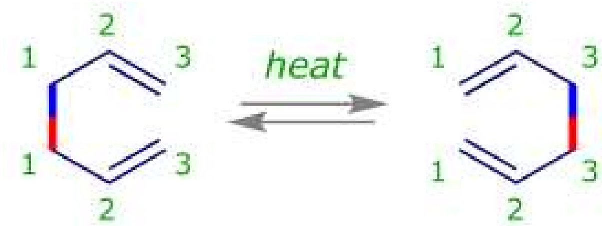
## *SIGMATROPIC REARRANGEMENTS*

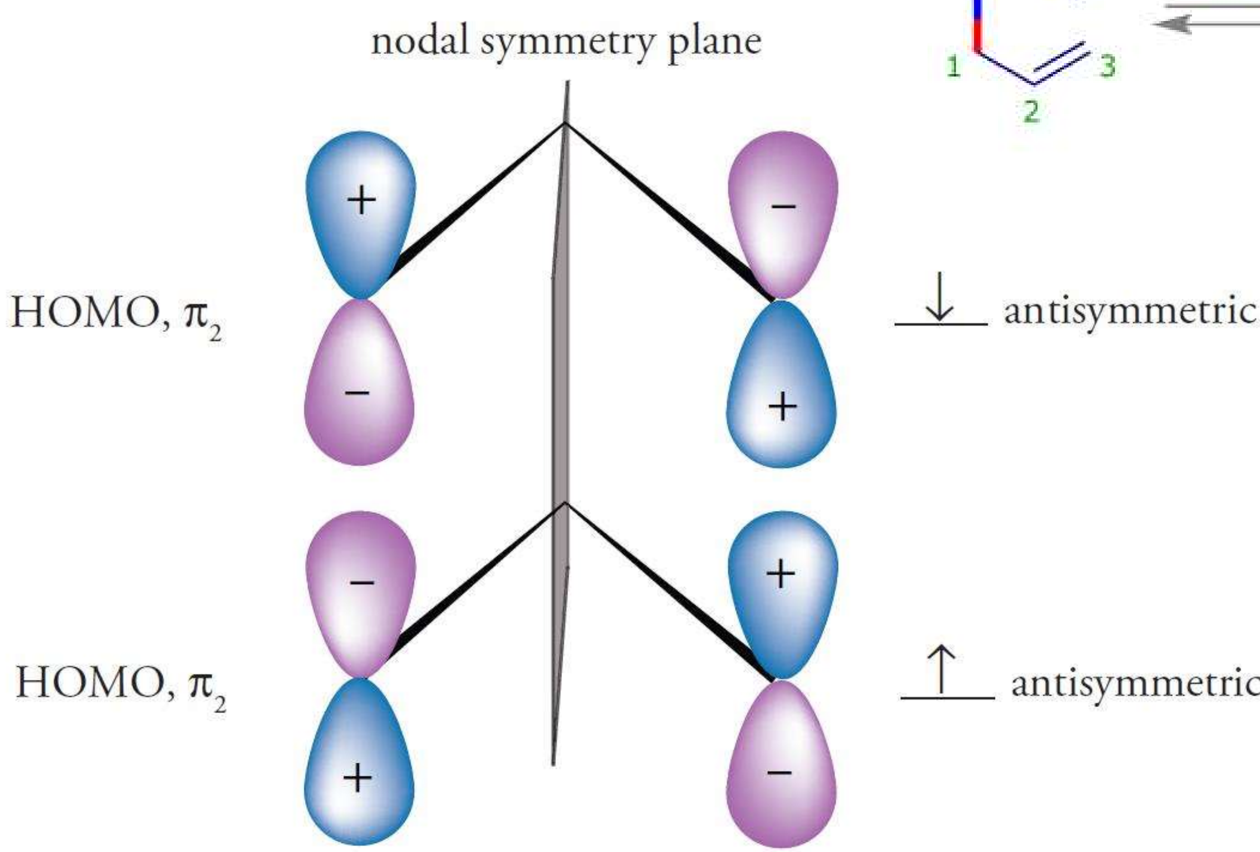
| Electrons in TS | Activation | Symmetry allowed |
|---|---|---|
| **4n** (even number of curved arrows) | heat | antara* |
| | light | supra |
| **4n + 2** (odd number of curved arrows) | heat | supra |
| | light | antara* |

* *In most cases restrictions imposed by molecular geometry prevent antarafacial reactions.*

# [3,3] Sigmatropic Rearrangements

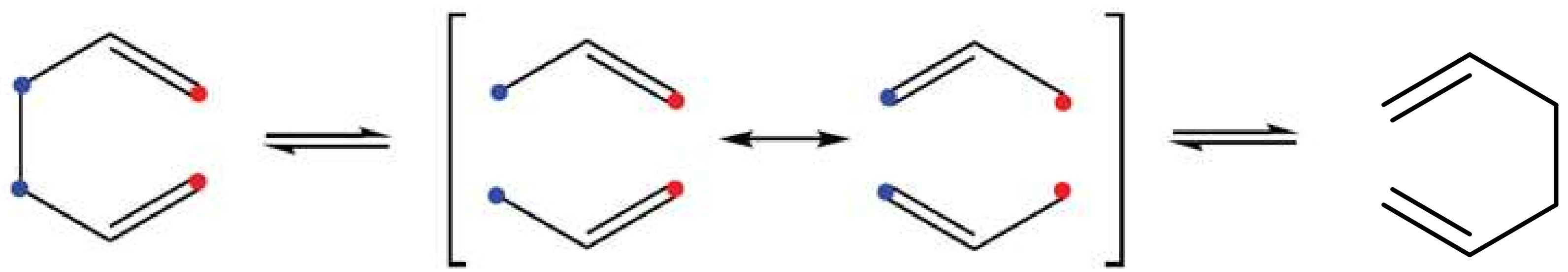
A [3,3] sigmatropic rearrangement such as the Cope rearrangement of 1,5-hexadiene can be thought of as a process in which two three-carbon radical fragments form and then bond again in a different position. Again, as we noted previously, the reaction does not proceed by free radical intermediates. However, the same orbitals are involved in the concerted process. So we can explain the experimental results using this model



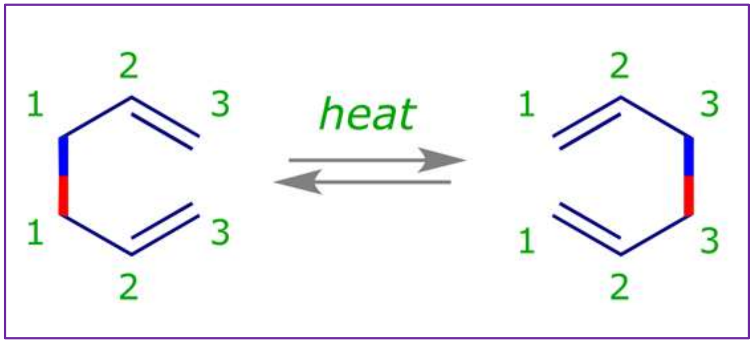
# [3,3] Sigmatropic Rearrangements

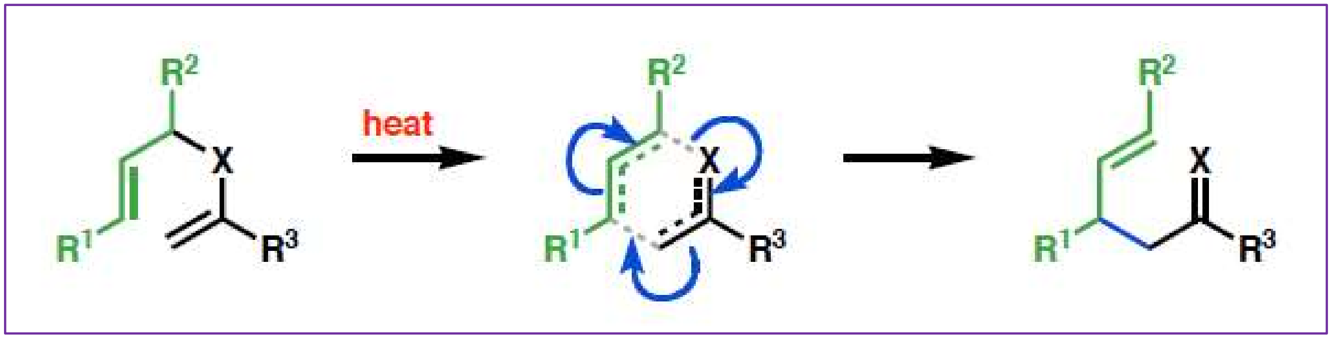
The resonance forms of the two allyl radicals used as models for the reactions are shown as a transition state in the following equation.



The resonance forms of the two allyl radicals used as models for the reactions are shown as a transition state in the following equation. An allyl radical contains three $\pi$ electrons, two in $\pi_1$ and one in the $\pi_2$ molecular orbital. Therefore, the frontier molecular orbital is $\pi_2$. It is antisymmetric. The bond that is broken must generate 2p atomic orbitals with lobes of the same sign directed toward each other. The bond to be formed must occur between 2p atomic orbitals with lobes of the same sign directed toward each other. This can be accomplished using the HOMO of one radical to form a $\sigma$ bond with the HOMO of the other radical.

# 3,3-Sigmatropic Rearrangement

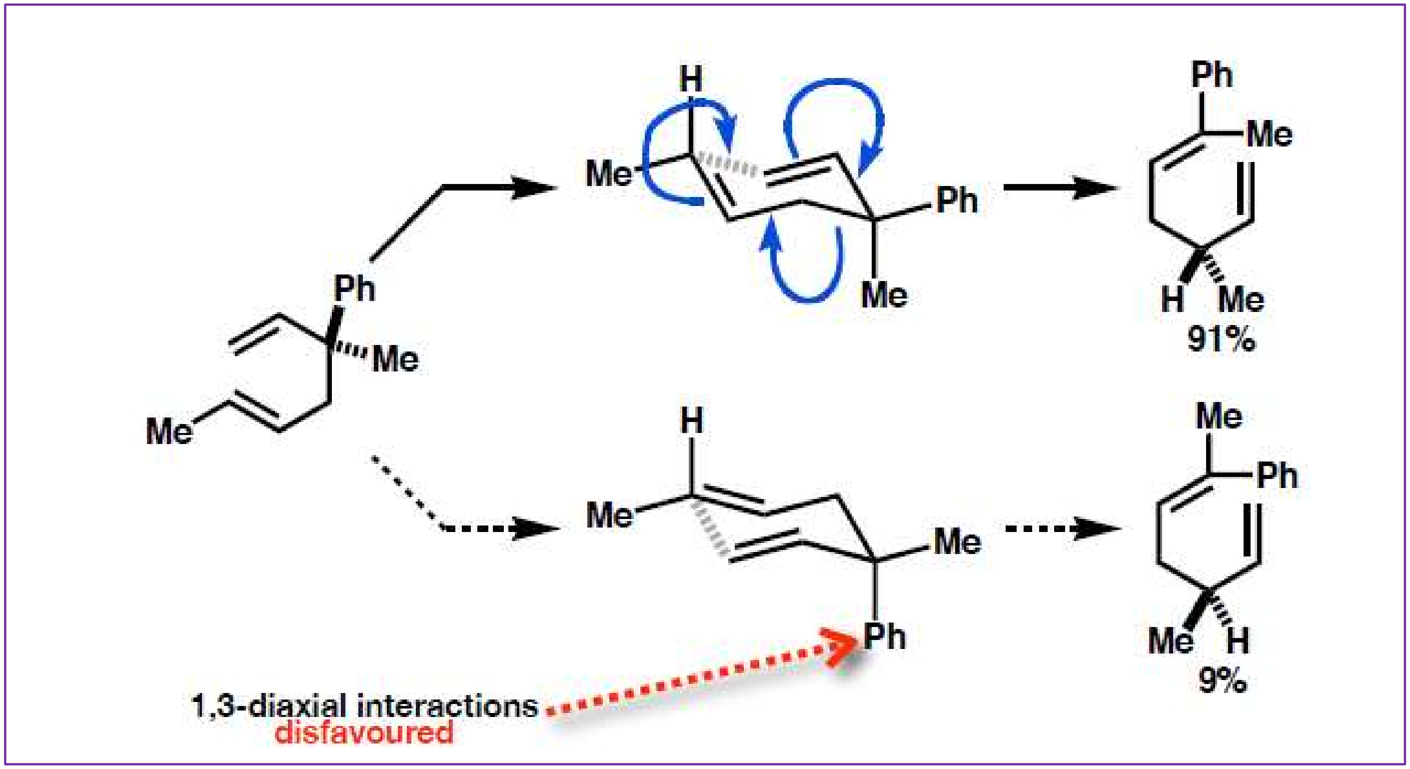



- A class of **pericyclic reactions** whose stereochemical outcome is governed by the **geometric** requirements of the cyclic transition state
- Reactions generally proceed *via* a **chair-like** transition state in which **1,3-diaxial interactions** are minimised

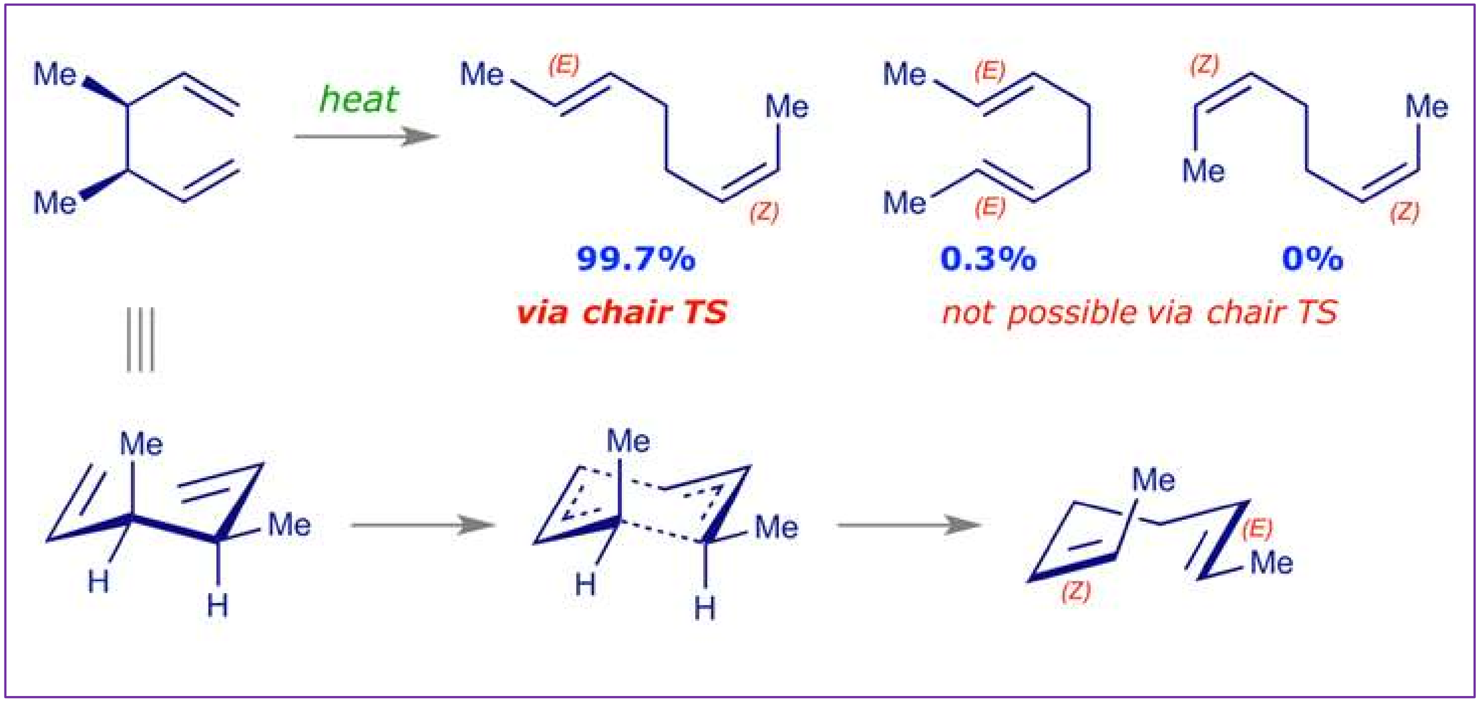
**Absolute stereochemistry** - controlled by existing stereocentre (destroyed in rct)
**Relative stereochemistry** - controlled by alkene / enolate **geometry**

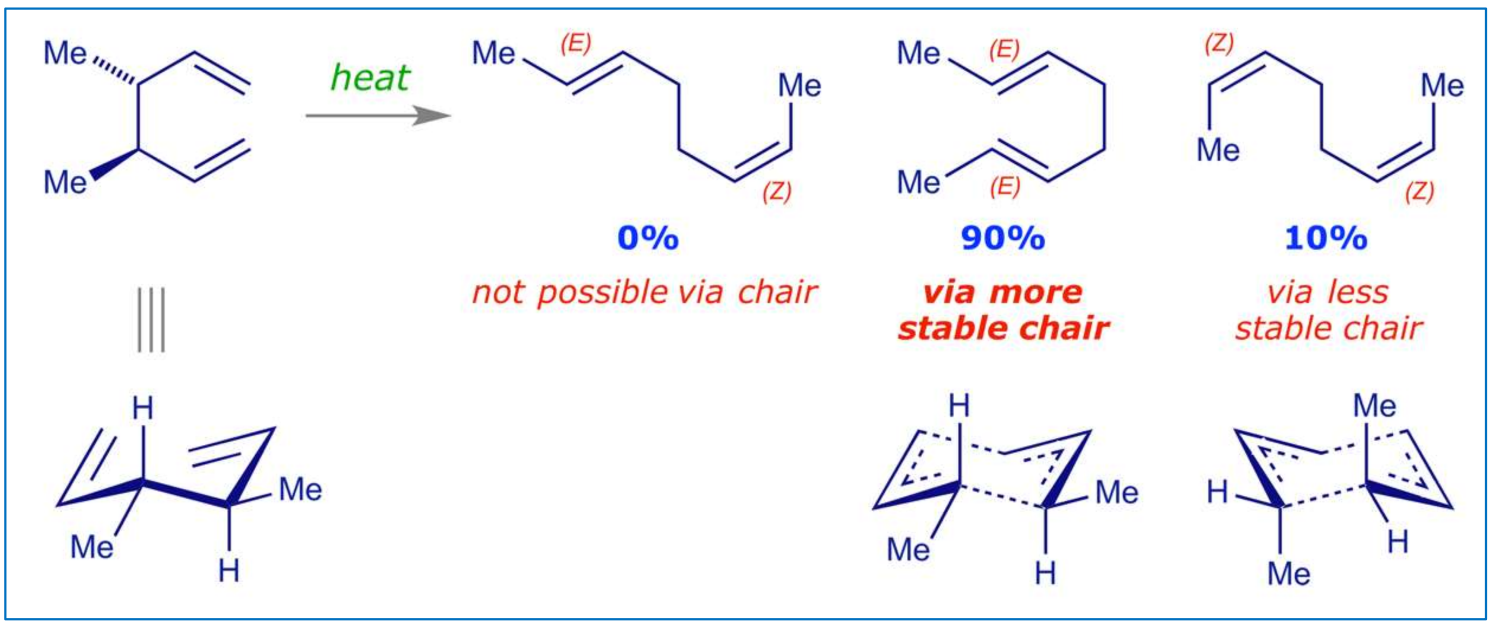
# Cope Rearrangement




- A very simple example of a substrate controlled [3,3]-sigmatropic rearrangement is the **Cope** rearrangement
- To minimise **1,3-diaxial** interactions **phenyl group** is **pseudo-equatorial**
- Note: the original stereocentre is **destroyed** as the new centre is formed
- This process is often called **'chirality transfer'**

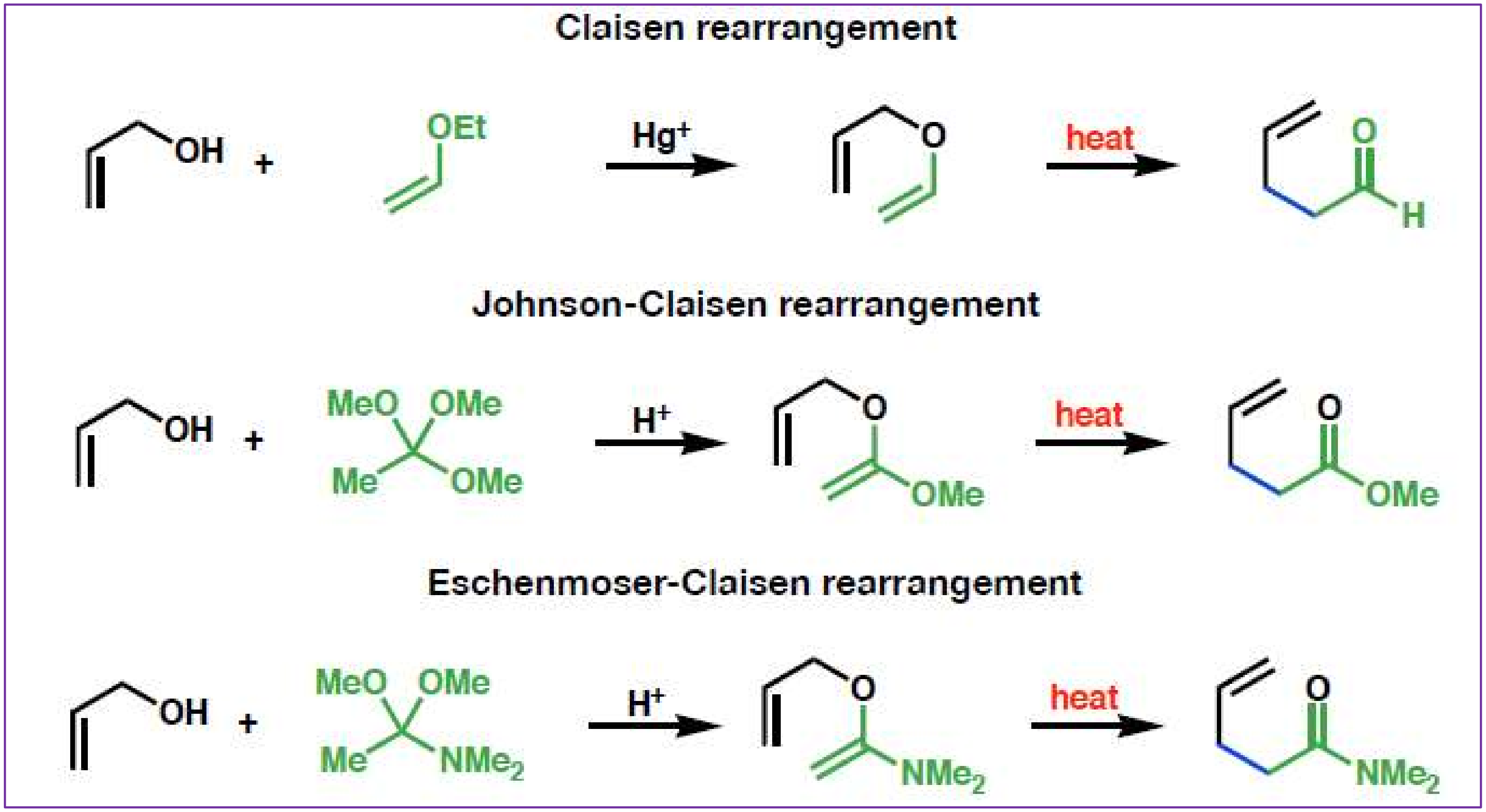
# Cope Rearrangement



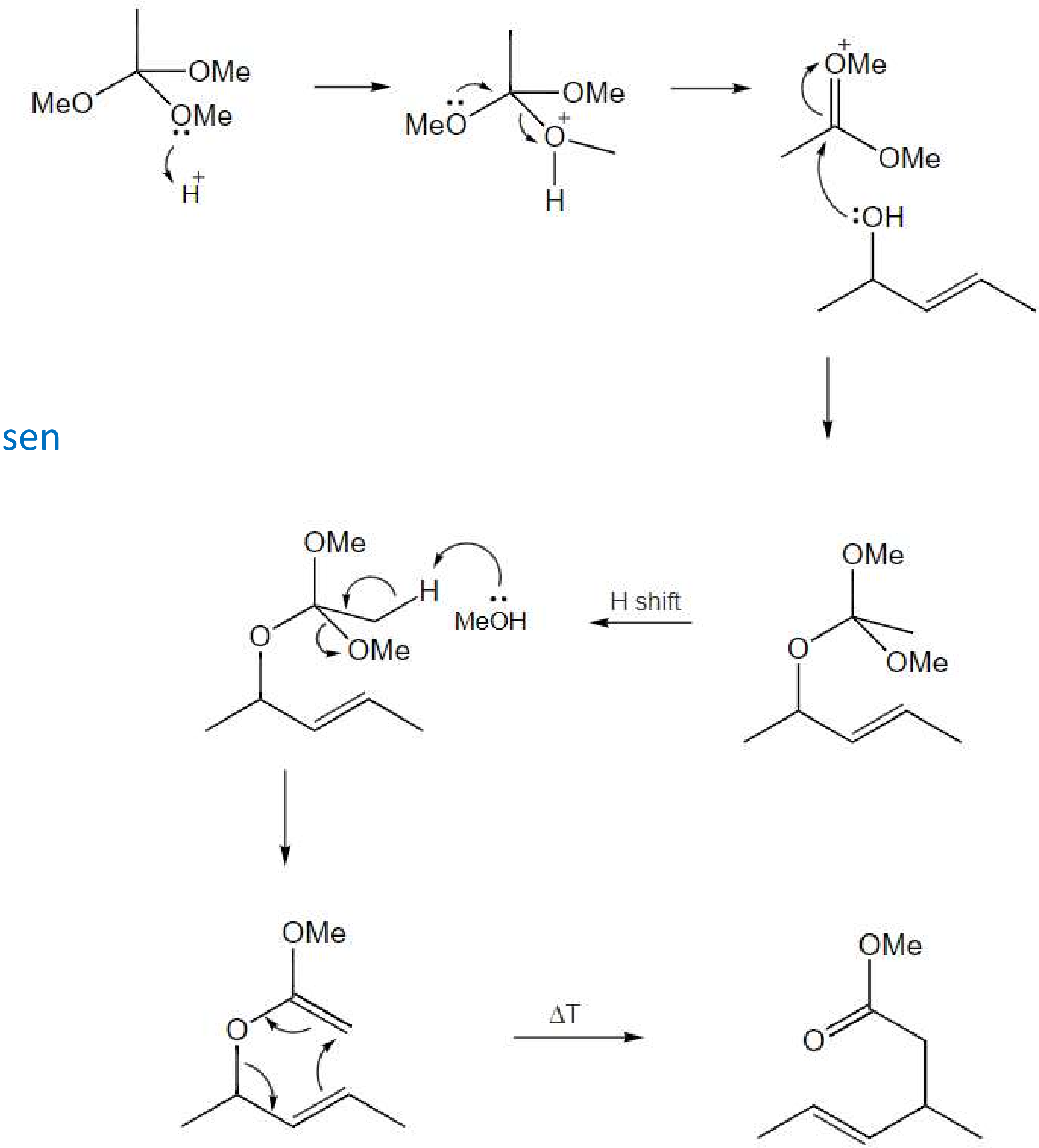



# Claisen Rearrangements

The Claisen rearrangement (not to be confused with the Claisen condensation) is a powerful carbon–carbon bond-forming chemical reaction discovered by Rainer Ludwig Claisen in 1912. The heating of an allyl vinyl ether will initiate a [3,3]-sigmatropic rearrangement to give a γ,δ-unsaturated carbonyl.

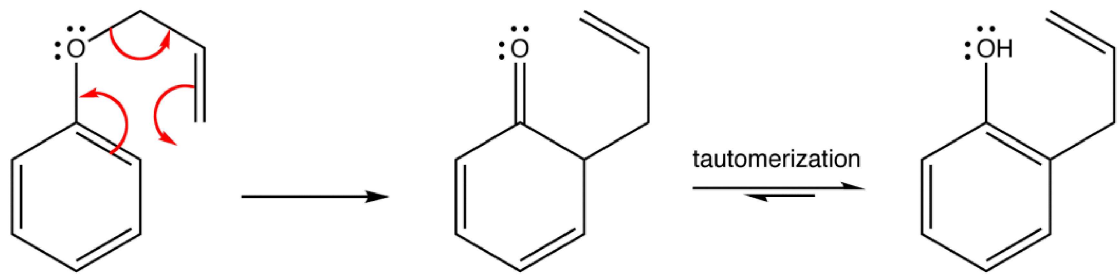


Johnson-Claisen



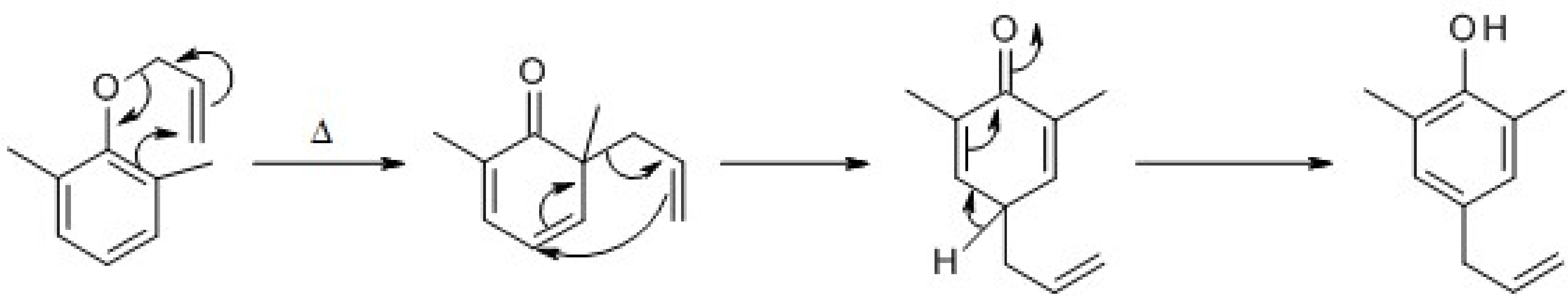
# Aromatic-Claisen Rearrangement

## ortho-Claisen

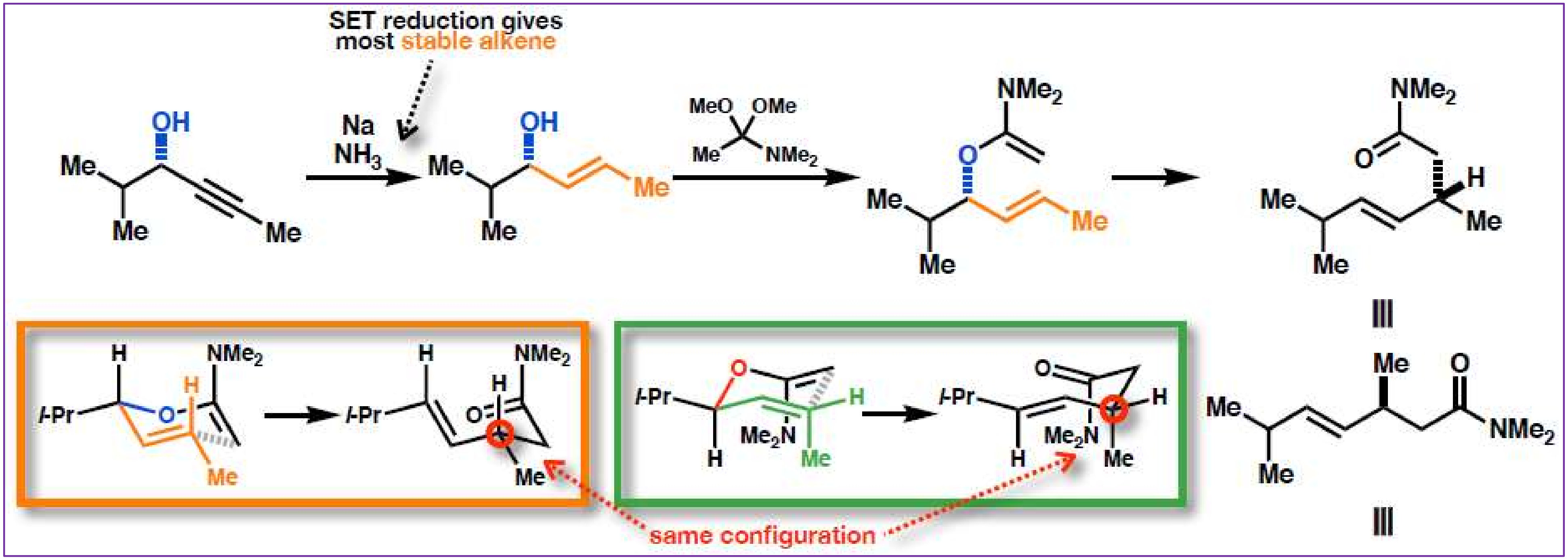


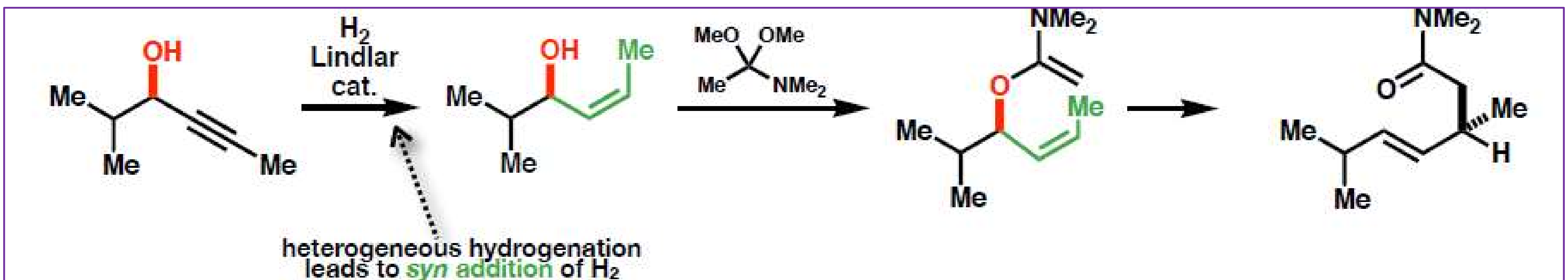

Allyl-phenyl ether

## para-Claisen



# Enantioconvergent Eschenmoser-Claisen Synthesis




- Both enantiomers of initial alcohol can be converted into the **same** enantiomer of product
- This process (Eschenmoser-Claisen) shows the importance of **alkene geometry**